Valentina Sanchez Melchor

+(31) 616352615 | valentinasmtr333@gmail.com | GitHub

PhD candidate in Machine Learning at Tilburg University, researching graph and topological approaches to representation learning, with work presented at ICML and NeurIPS workshops.

Education

Ph.D., Machine Learning (Representation Learning) — Tilburg University, NL
Department of Cognitive Science and Artificial Intelligence — *Oct. 2023 – Oct. 2027*

- Developing scalable software for representation learning in complex network structures with applications to large-scale datasets.
- Advancing Graph Neural Network (GNN), Topological Deep Learning (TDL) architectures for improved performance and efficiency.
- Researching novel representation learning approaches for deep learning architectures with applications in computational neuroscience, fMRI data and time-series data.
- Contributed a lifting and a model integration to TopoBench, the Geometric Intelligence Lab's (UCSB) framework; which I will join as a visiting researcher (Jan–Mar 2027).

MSc Data Science and Artificial Intelligence — Queen Mary University of London, UK
Sept. 2022 – Sept. 2023

- Achieved Distinction with an overall score of 85%.
- Dissertation - Semi-Supervised Node Classification with Graph Neural Networks: Leveraging Dual-Label Training and Community Detection-Based Features, achieving a score of 82%.
- Relevant modules: Machine Learning, Data Structures and Algorithms, Neural Networks and Deep Learning, Risk and Decision-Making for Data Science and AI, Programming for AI and Data Science, Information Retrieval, Data Mining, Ethics Regulation and Law.

BA Philosophy and Mathematics with a Year in Data Science — University of Kent, UK
Sept. 2018 – June. 2022

- Achieved 2:1 with an overall score of 75% on the Year in Data Science.
- Project - Assessing the Correlation Between Prison Staff and Inmate Mortality in Europe: A Seven-Year Study (2010-2017), achieving a score of 77%.
- Relevant Modules: Big Data and other Analytical Techniques, Predictive Modelling, Logic, Linear Algebra, Calculus, Graph Theory.

Projects

Current Research – The Human Brain as a Combinatorial Complex

- Developing an information-theoretic framework for constructing **Combinatorial Complexes (CCs)** from multivariate time-series that capture pairwise and higher-order dependencies under a unified data structure.
- Modelling higher-order dynamics using an ODE to establish when and how higher-order topological structures are present in multivariate data and can be captured by existing detection metrics.
- Implementing a topological deep learning (TDL) framework to capture and exploit higher-order dependencies in complex systems.

MSc Project – Graph Neural Networks for Semi-Supervised Learning

- Developed semi-supervised GNNs for node classification on `ogbn-arxiv` using dual-label training and community-informed features.
- Reached **87%** accuracy with community-derived labels vs. **72%** with original classes, showing the benefit of structural supervision.
- Aligned label spaces with graph topology to improve generalization in semi-supervised settings.

Technical Projects | Link

- Implemented a TopoBench-native adaptation of SheafHyperGNN (linear, diagonal variant) from Duta et al., "Sheaf Hypergraph Networks" (NeurIPS 2023) for the Topological Deep Learning (TDL) Challenge 2026.
- Designed a 'topological lifting' method based on modularity maximization for the ICML TDL Challenge.
- Developed and optimized a 3-block CNN on CIFAR-10, achieving 81% validation accuracy via hyperparameter tuning and learning-rate scheduling.
- Built a voice-identification pipeline (90% validation accuracy) using spectral features and model comparison.
- Created a movie recommender using cosine similarity on the IMDb dataset.
- Developed a data downloader and pre-processor for the AOMIC-ID1000 dataset.

Publications

Peer-Reviewed Publications

- Data Augmentation Techniques for fMRI Data: A Technical Survey, **IEEE Access**.
- ICML Topological Deep Learning Challenge 2024: Beyond the Graph Domain, Proceedings of the Geometry-grounded Representation Learning and Generative Modeling Workshop (GRaM) at **ICML 2024**.
- TopoBench: A Framework for Benchmarking Topological Deep Learning, **DMLR**.
- A Generative Model for Higher-Order Network Dynamics, **JCN**. Under Submission.

arXiv

- The Human Brain as a Combinatorial Complex (2025). Accepted at the NeurReps Workshop **NeurIPS 2025**

Academic Activities

Selected Conferences & Workshops

- **ICML GRaM Workshop** | Link - Honorable mention, Topological Deep Learning challenge. *2024*
- ECAI Doctoral Consortium - Talk & poster presentation on 'Topological Deep Learning for Brain Network Analysis'.
- Oxford Machine Learning Summer School (**OxML**) - Representation learning program.
- Panelist at the 'AI and the Researcher of Tomorrow' event, Tilburg University. *2025*
- **NetSci Conference** - Presentation 'The Human Brain as a Combinatorial Complex' at the main conference and at the HONAI satellite.
- AIM (AI and Mathematics) Workshop.
- **NeurIPS NeurReps Workshop** - Poster presentation on 'The Human Brain as a Combinatorial Complex'.
- Machine Learning Summer School (**MLSS**) NYC 2026, **Columbia University**. *2026*

Academic & Teaching Service

- Teaching Assistant for **Deep Learning** and **Data Mining** (MSc level), Tilburg University. *2024*
- Master's Thesis Grader, Department of Cognitive Science and Artificial Intelligence, Tilburg University.
- Reviewer for the **NeurReps Workshop** at NeurIPS. *2025 & 2026*

Technical Skills

Languages: Python, SQL, R.
Tools: Git/GitHub, VSCode, PyCharm, Docker, AWS.

Work Experience

Lead Generation Representative July 2022 – Sept. 2022
Lavanda *London, UK*

- Generated and managed a database of over 30,000 leads using advanced Excel skills.
- Utilized Excel pivot tables, macros and VLOOKUPs for data segmentation and analysis, providing insights into target markets.
- Programmed with VBA to perform customized calculations, aiding strategic planning.

Data Analysis Industry Training June 2022 – July 2022
Brainnest *Bremen, Germany*

- Used SQL to query, clean, and analyze large-scale datasets, supporting data-driven decision-making.
- Employed multiple regression techniques including logistic regression and linear regression, utilizing p-values for variable selection and ensuring statistical significance.
- Developed regression models with an average prediction accuracy of 80%, validated through statistical significance testing.

Languages and Interests

- Languages: English (C2/bilingual), Spanish (native).
- Interests: singing, fitness, neuroscience.